Benchmark design: five controllers, one deterministic testbed

Step 1: policy training

training weather 2014–2021

PPO, 5 seeds, reference recipe

checkpoints at 6 budgets,

50k to 2M environment steps

arms 1 and 3 require no training

Controller arms

1. rule-based incumbent (no model)
2. black-box RL (PPO)
3. economic NMPC (physics only)
4. RL-in-MPC (physics-informed)

Step 2: evaluation grid

held-out weather 2022–2025,

model mismatch 0 / 5 / 10%,

5 seeds, 60-day seasons

284 independent seasons

per-arm axes: 4 + 20 + 60 + 60;

mismatch applies to arms 3 and 4

GreenLight-Gym2, tomato

28-state greenhouse ODE, 15-minute steps

deterministic plant, true parameters θ

controllers receive a perturbed nominal θ only

Season-level scoring

net economic value [EUR/m²]

humidity feasibility gate

Clopper–Pearson; crash counts

online cost [ms/step]

solver reliability (arms 3, 4)

gate before ranking

Reproducibility

seeded end to end; repeat runs bit-identical within platform

each result carries its run record: commit, hardware, grid, failures